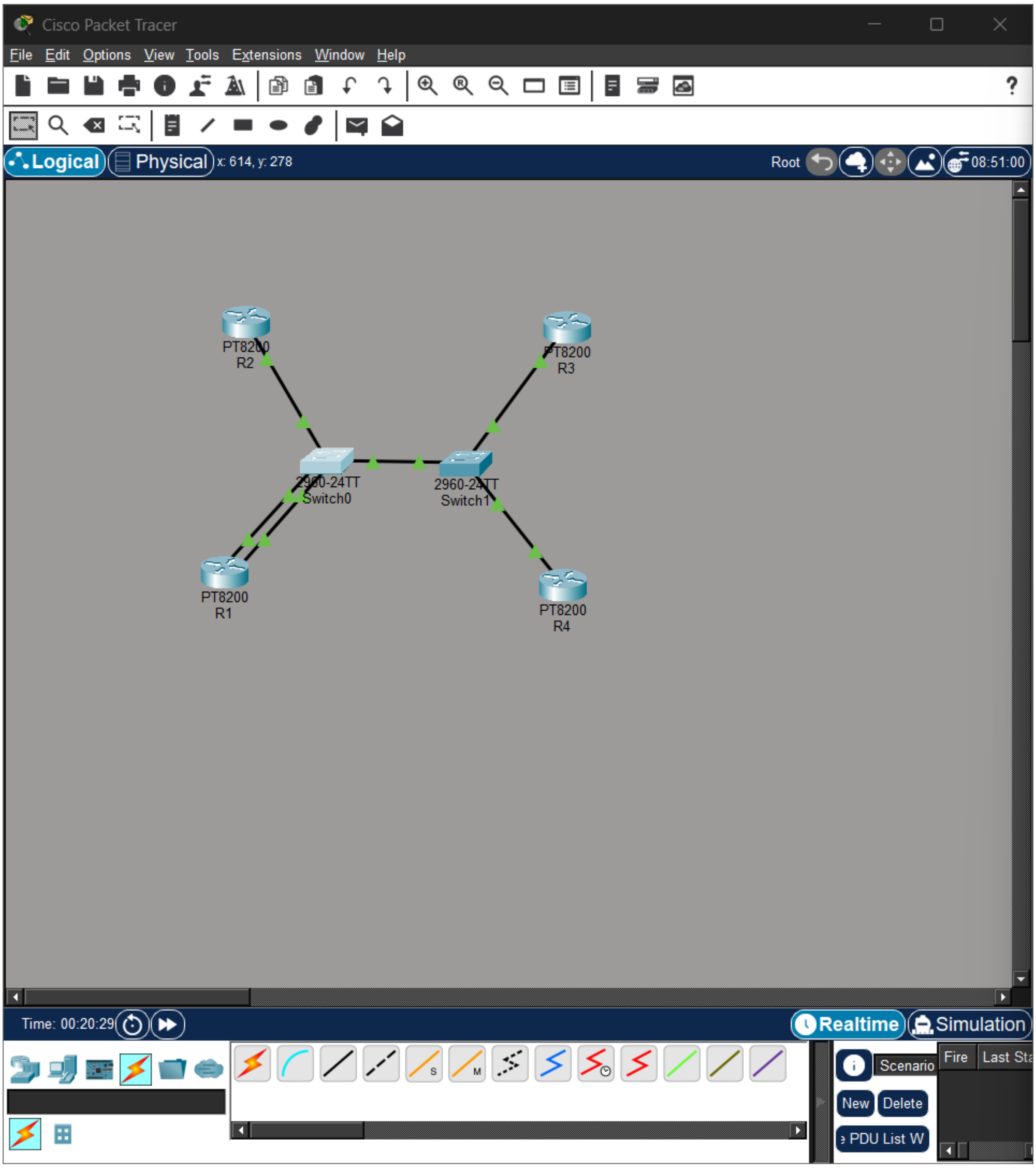
Ryno Badenhorst Week 2 Cisco



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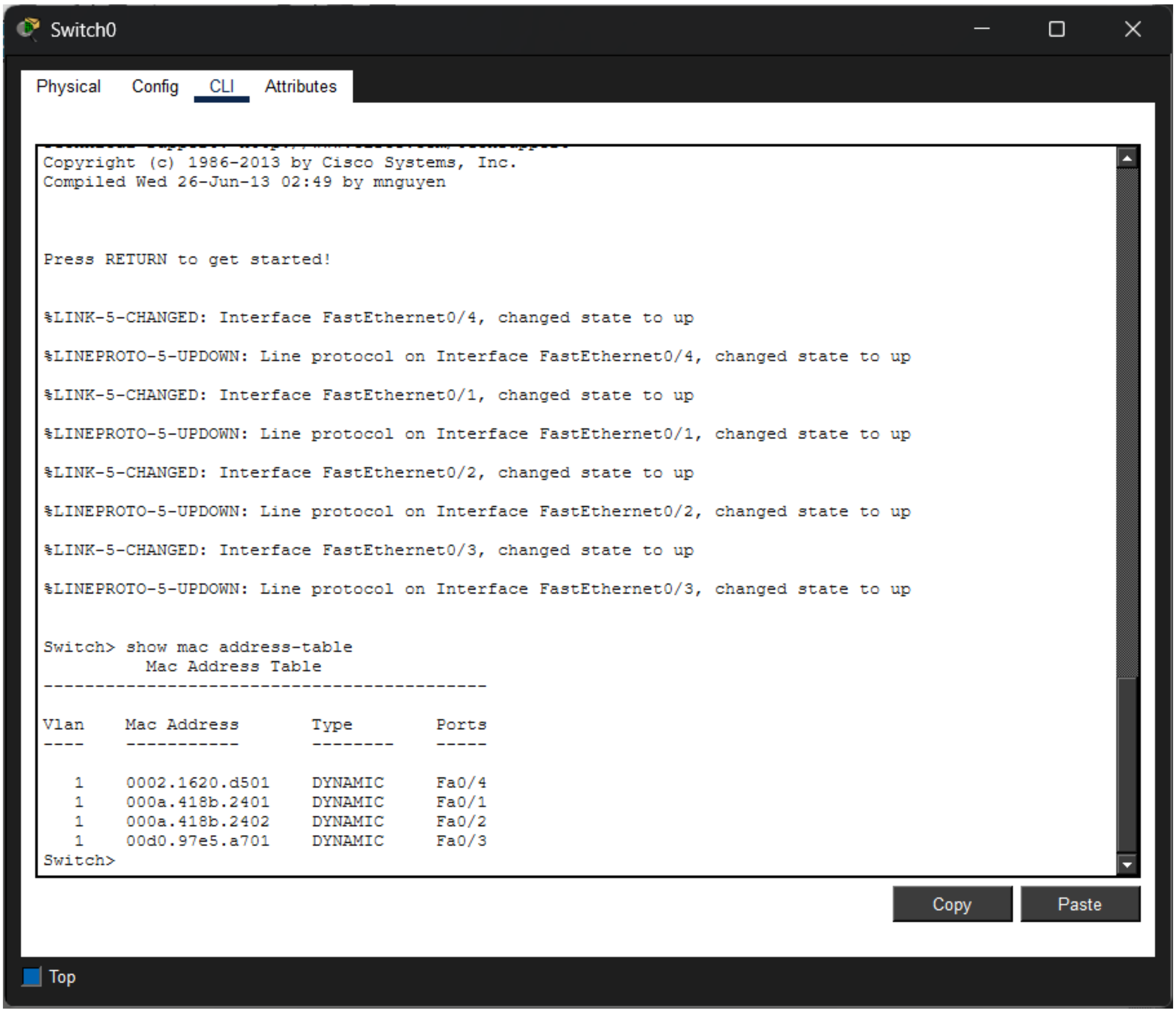
5.



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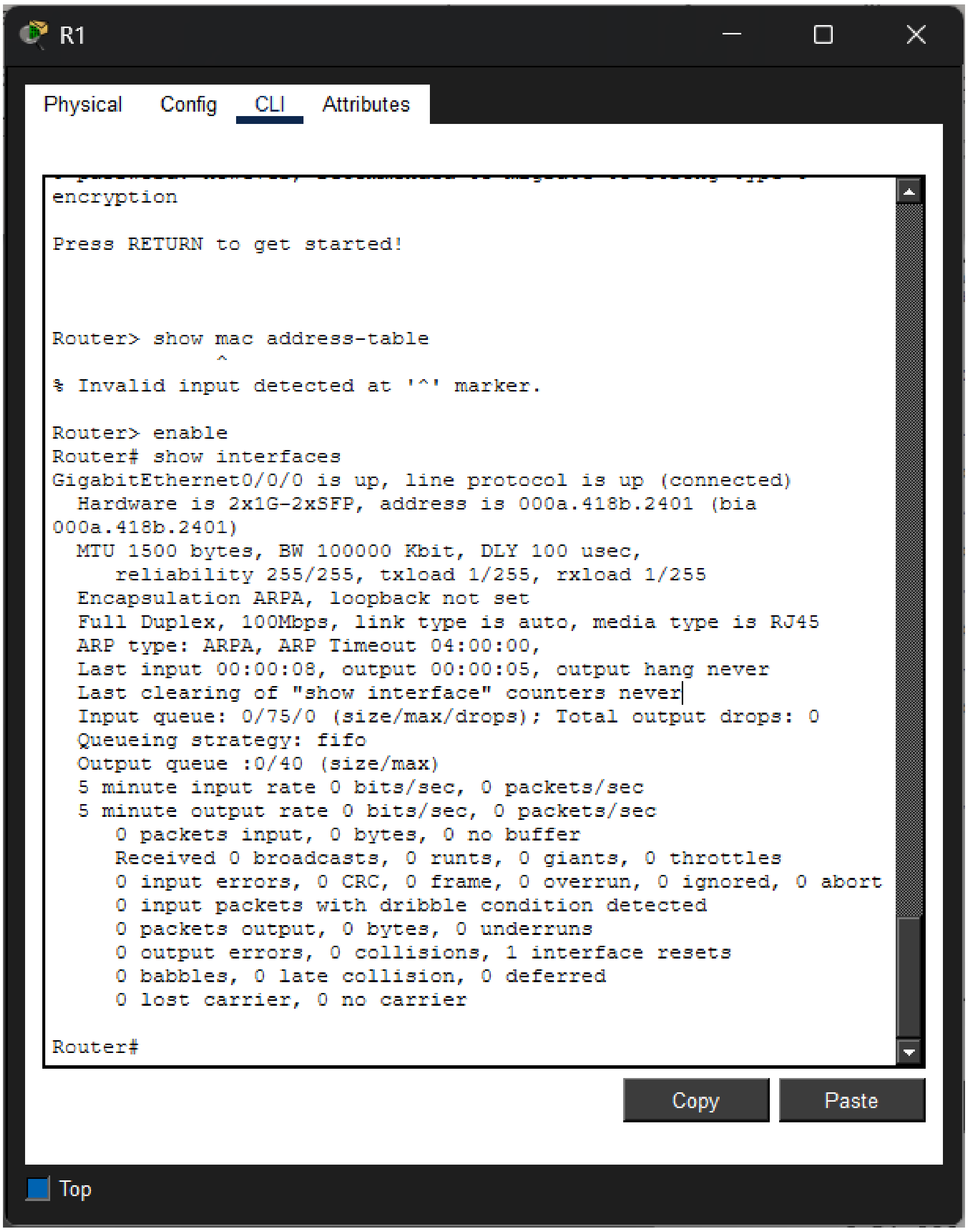
6. R1: 000a.418b.2401



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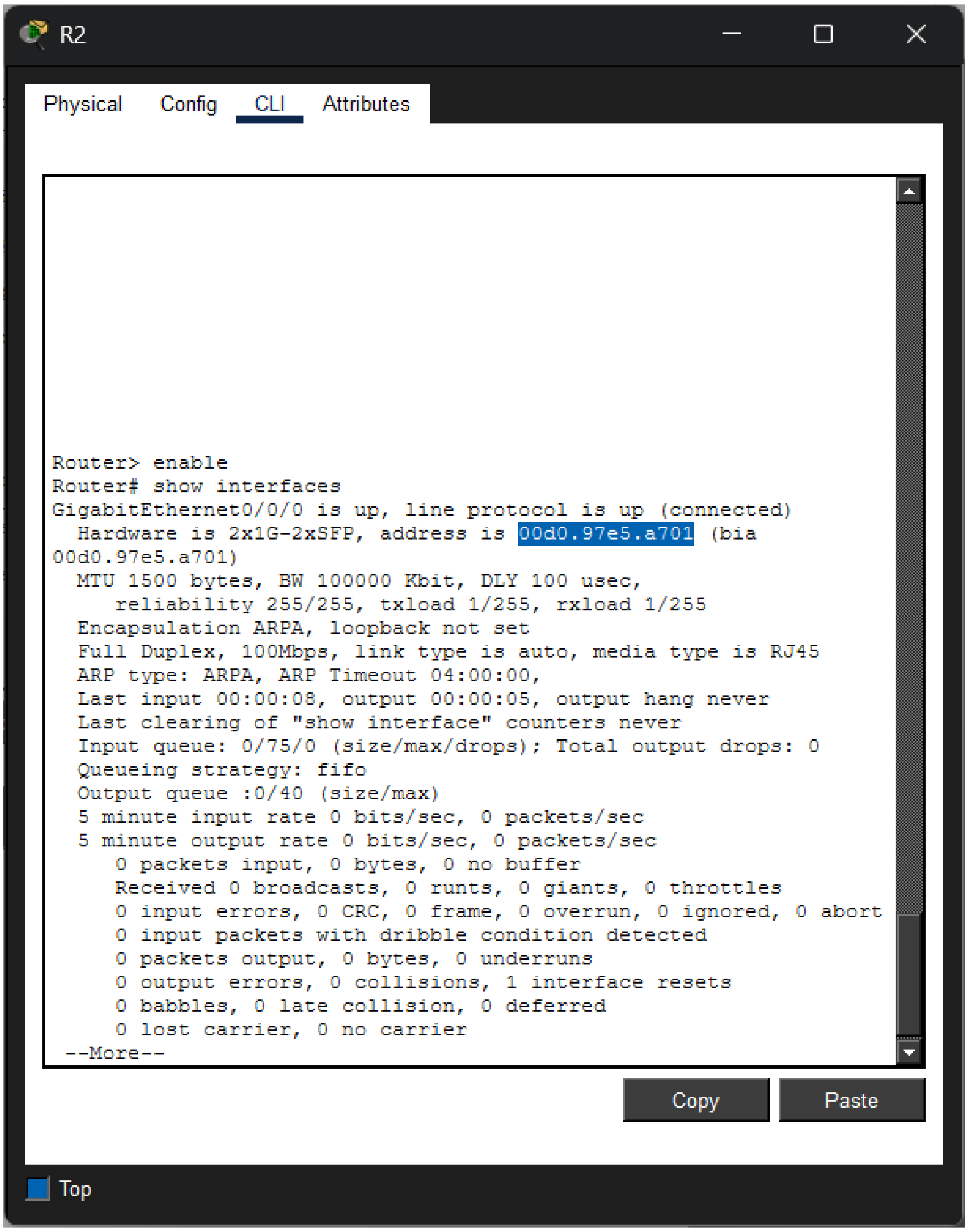
R2: 00d0.97e5.a701



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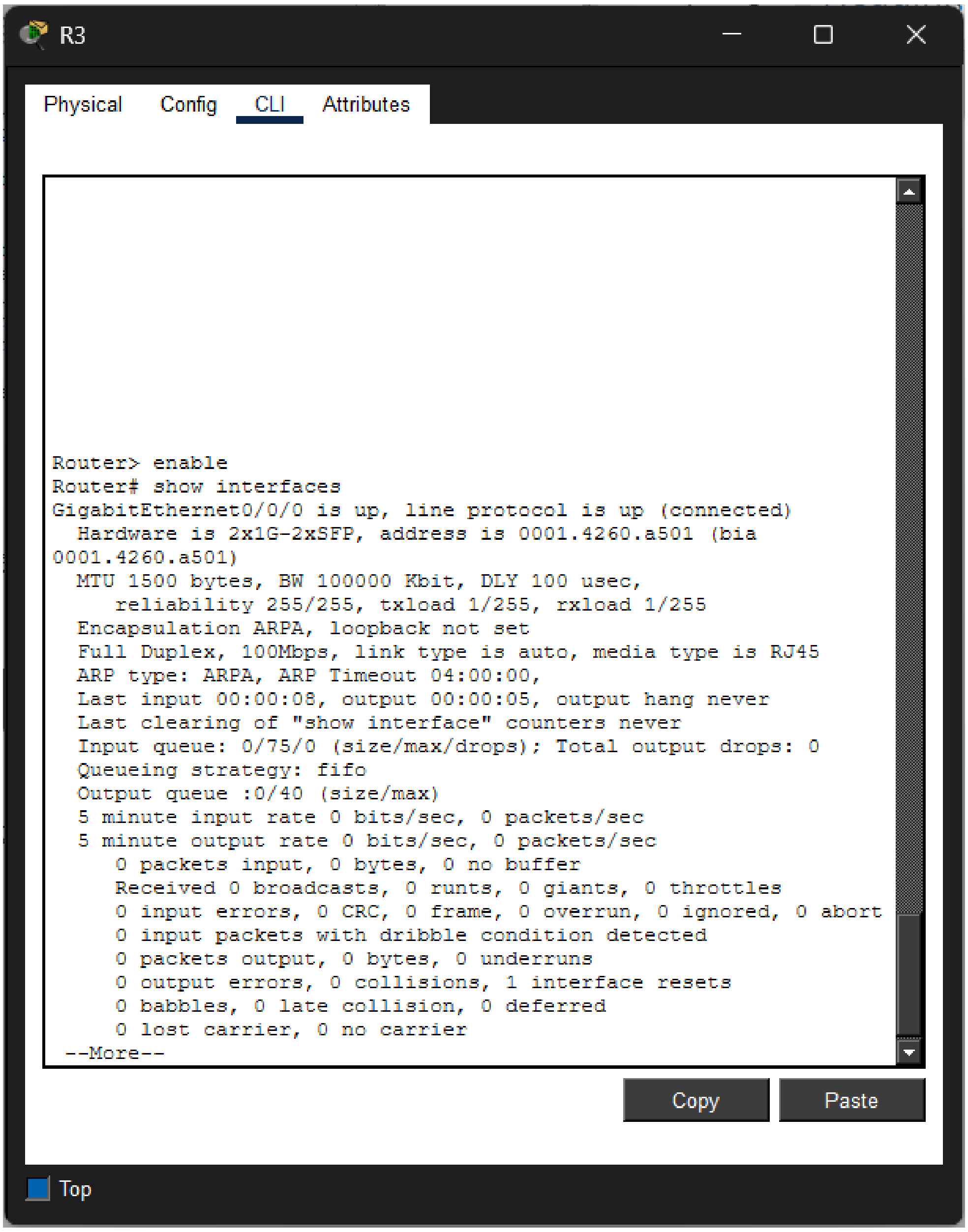
R3: 0001.4260.a501




```
Router> enable
Router# show interfaces
GigabitEthernet0/0/0 is up, line protocol is up (connected)
  Hardware is 2x1G-2xSFP, address is 0001.4260.a501 (bia
0001.4260.a501)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
 --More--
```

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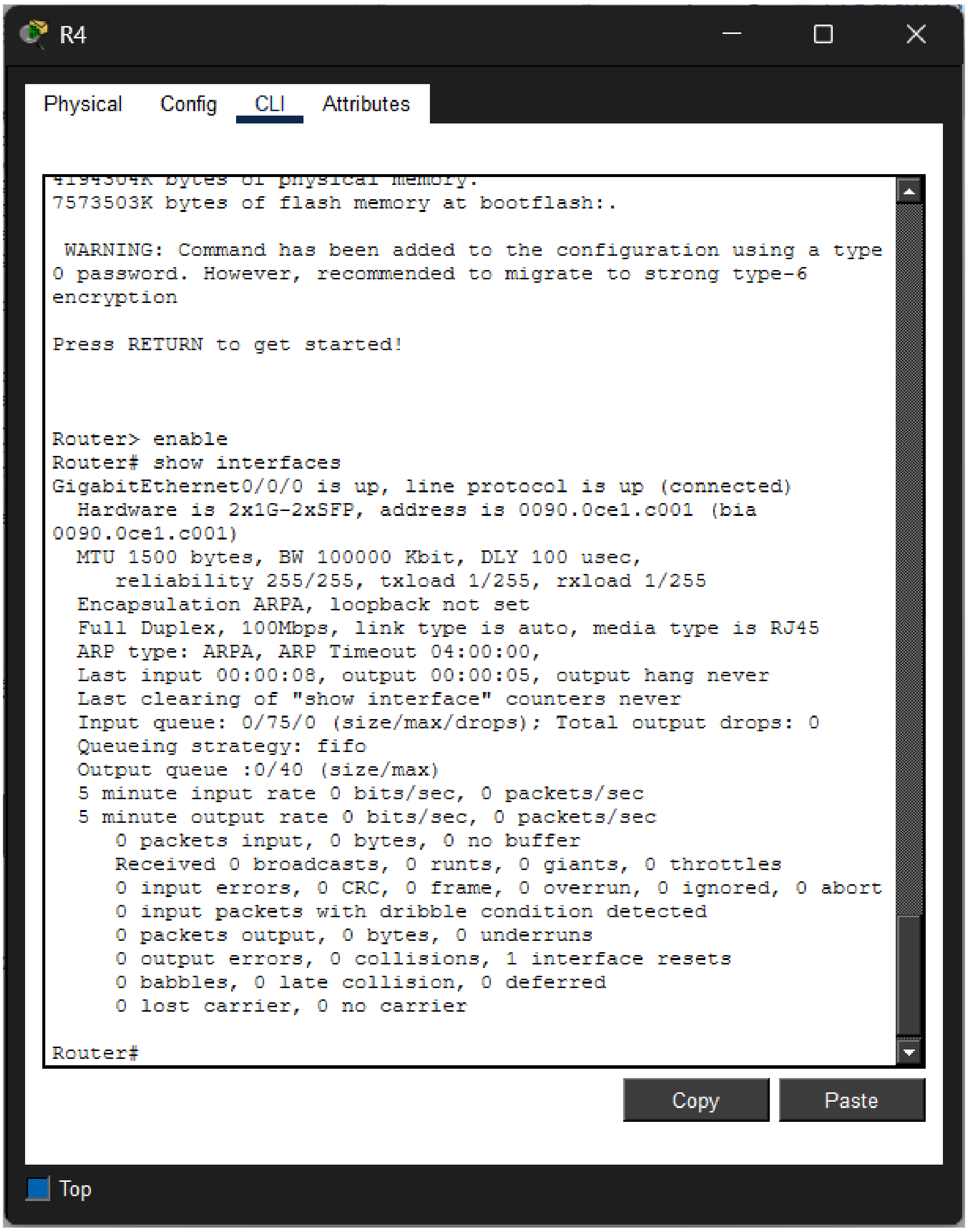
R4: 0090.0ce1.c001



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7. Ping from R1 to R2

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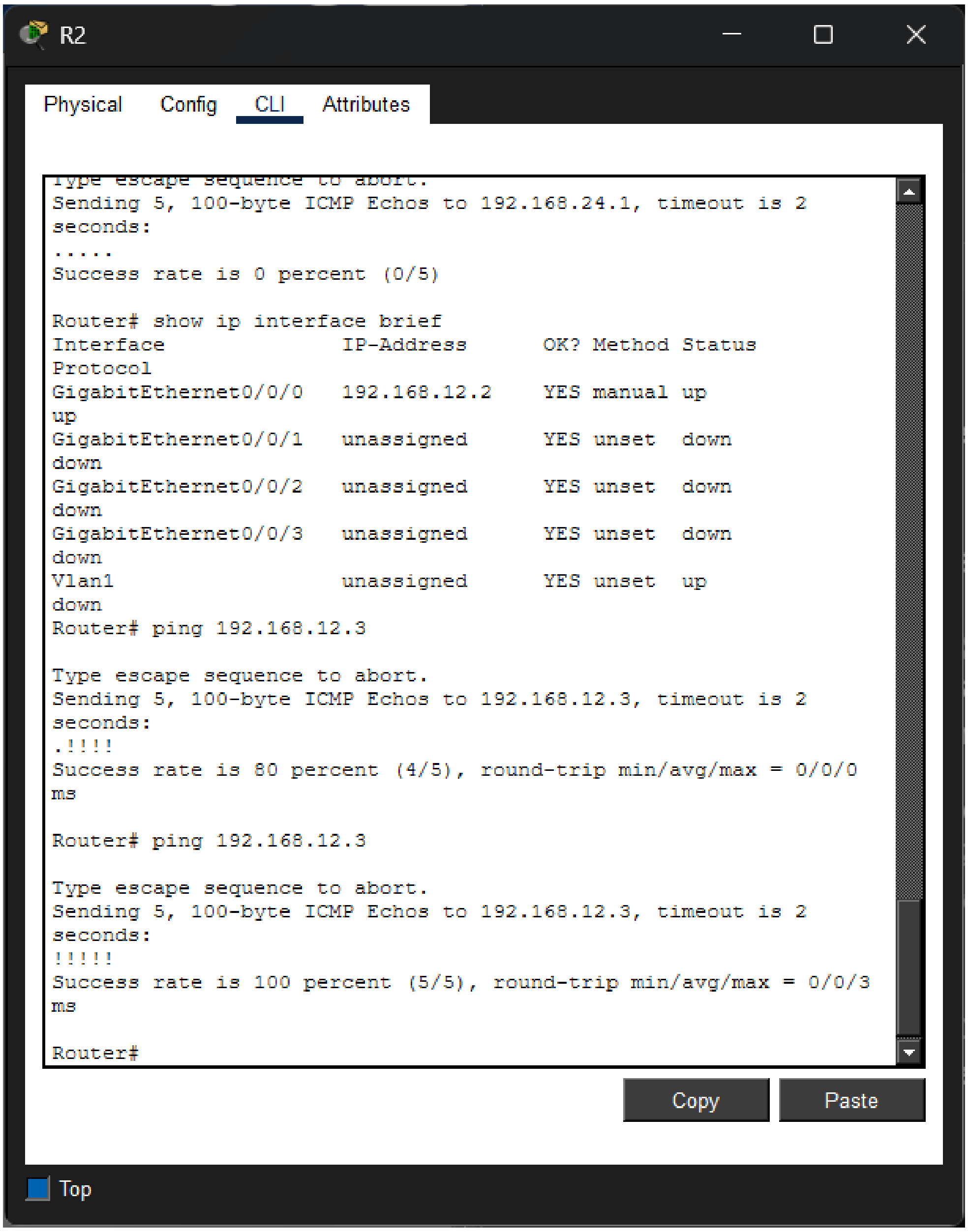
Ping from R2 to R3:




```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.24.1, timeout is 2
seconds:
.....
Success rate is 0 percent (0/5)

Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.2    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  down
down
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router# ping 192.168.12.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.3, timeout is 2
seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0
ms

Router# ping 192.168.12.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.3, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/3
ms

Router#
```

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Ping from R3 to R4:

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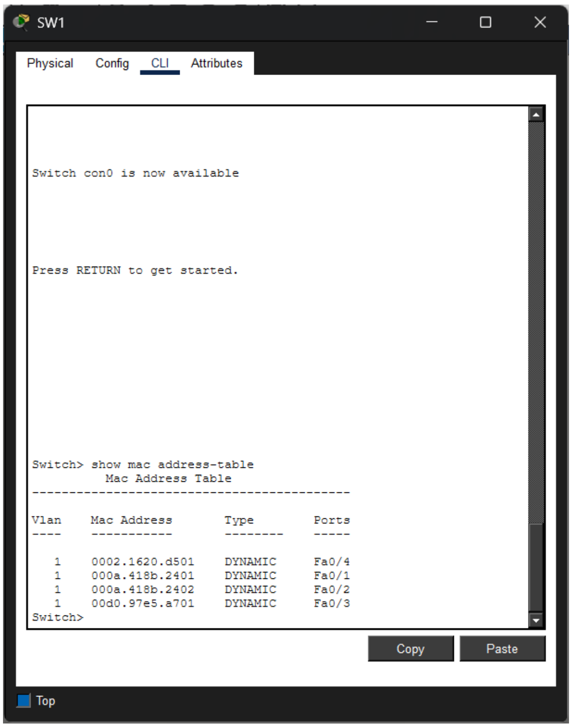
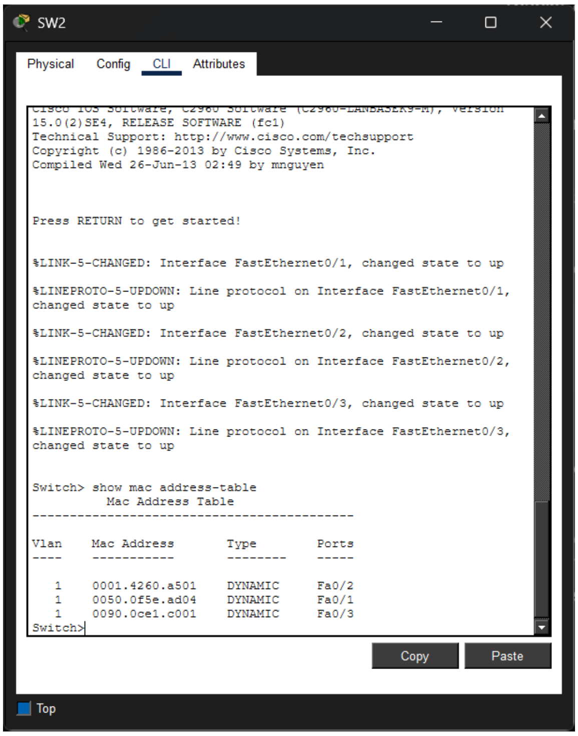
8.





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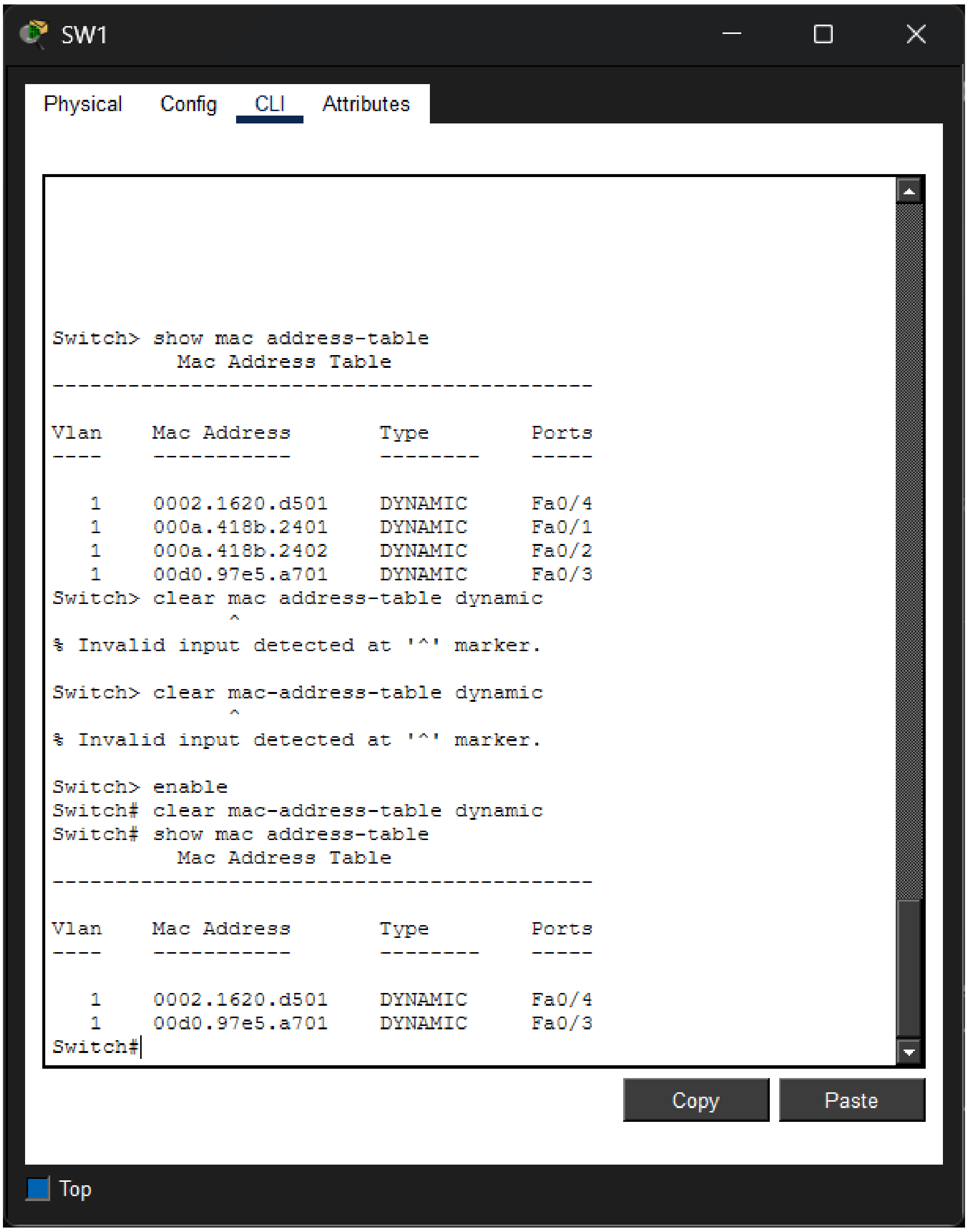
9. Clear Mac address and check again:

SW1:



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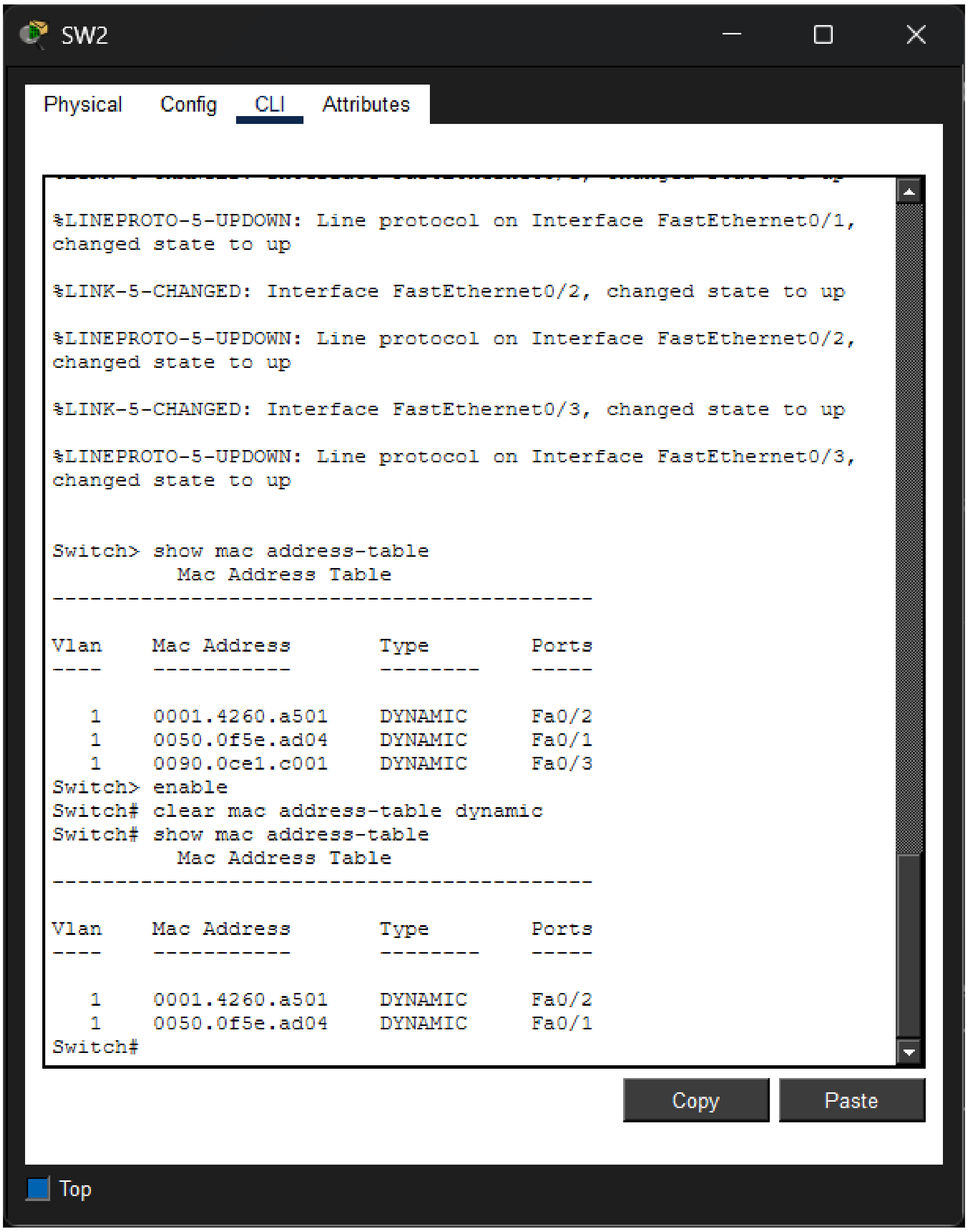
SW2:



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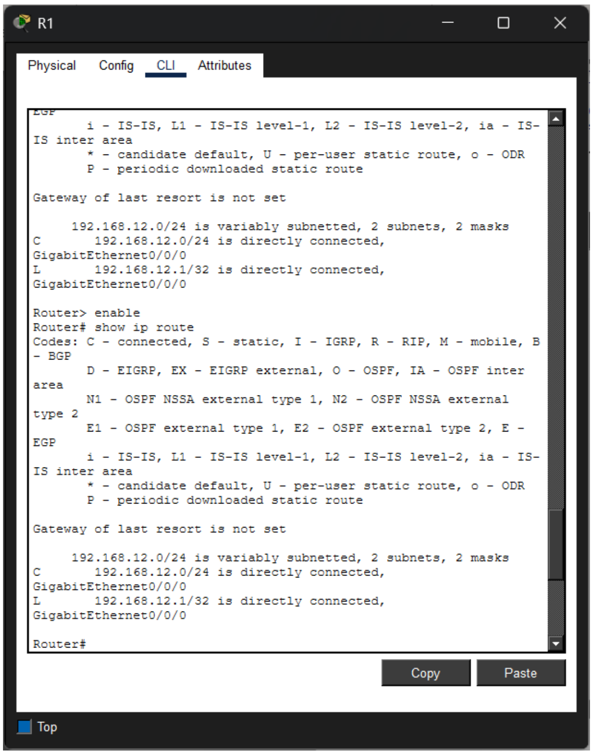
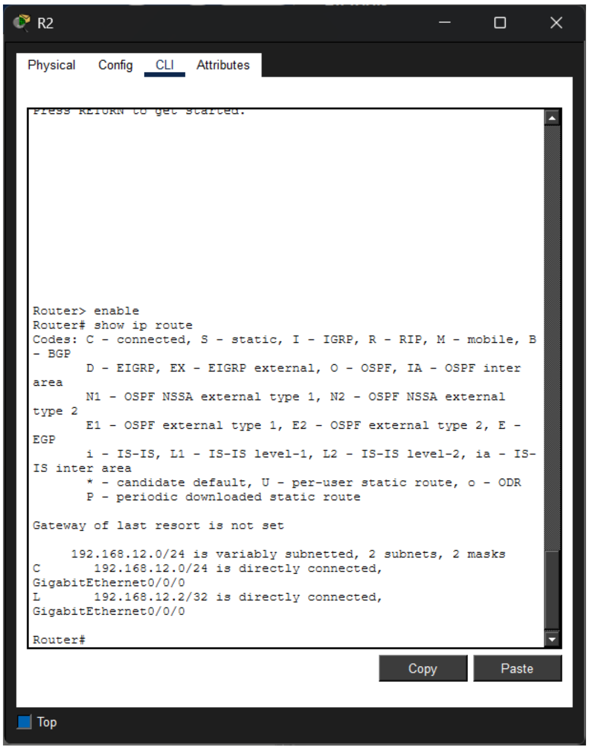
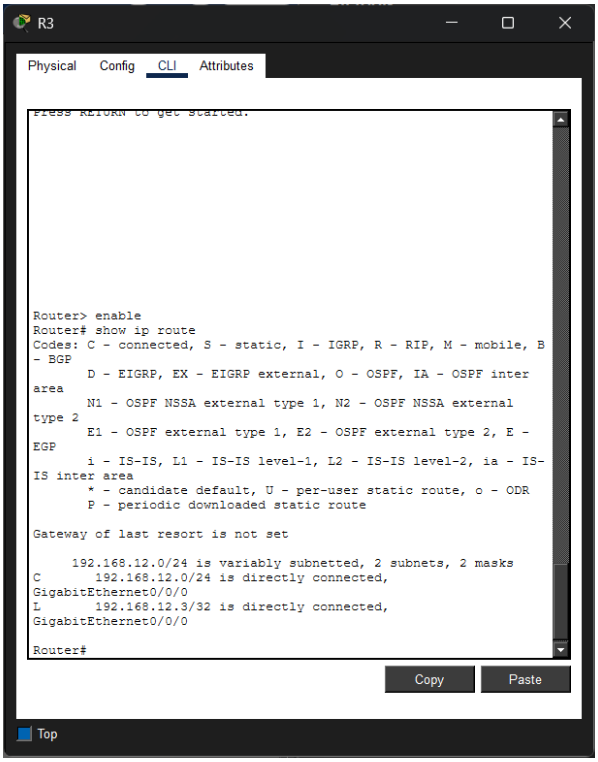
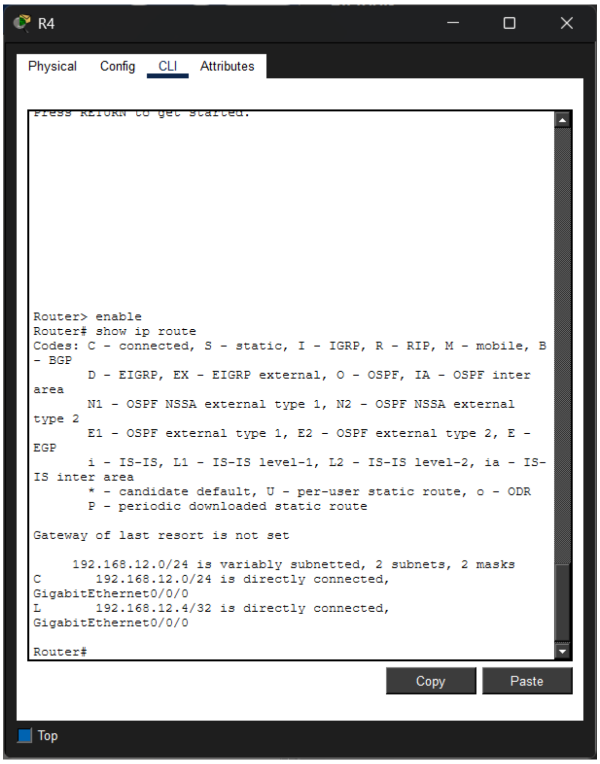
10.









Ryno Badenhorst Week 2 Cisco

11. Assigning IP addresses:

R1

Physical Config CLI Attributes

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:19:10.1919: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.1    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  up
up
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router#
```

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Top

R2

Physical Config CLI Attributes

```
IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.2/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:20:36.2020: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.2    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  down
down
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router#
```

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Top

R3

Physical Config CLI Attributes

```
IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.3/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.3 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:22:38.2222: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.3    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  down
down
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router#
```

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Top

R4

Physical Config CLI Attributes

```
IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected,
GigabitEthernet0/0/0
L       192.168.12.4/32 is directly connected,
GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.4 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:23:32.2323: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.4    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  down
down
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router#
```

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Top

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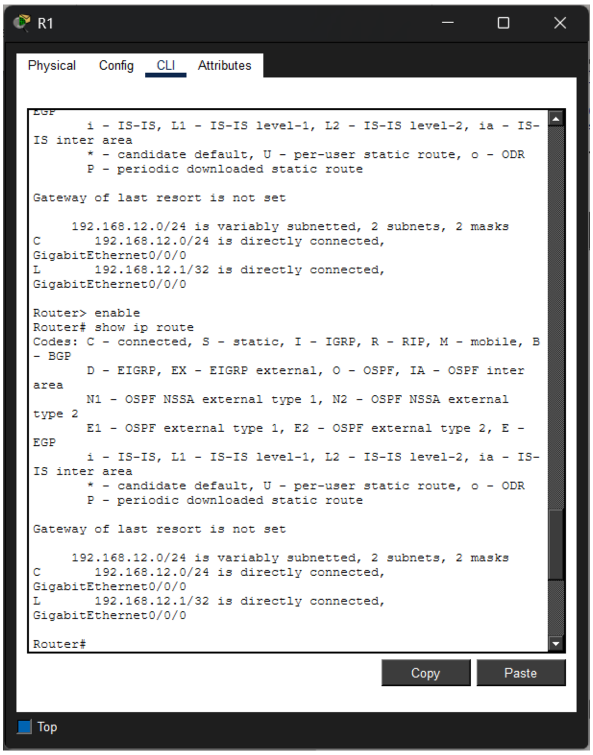
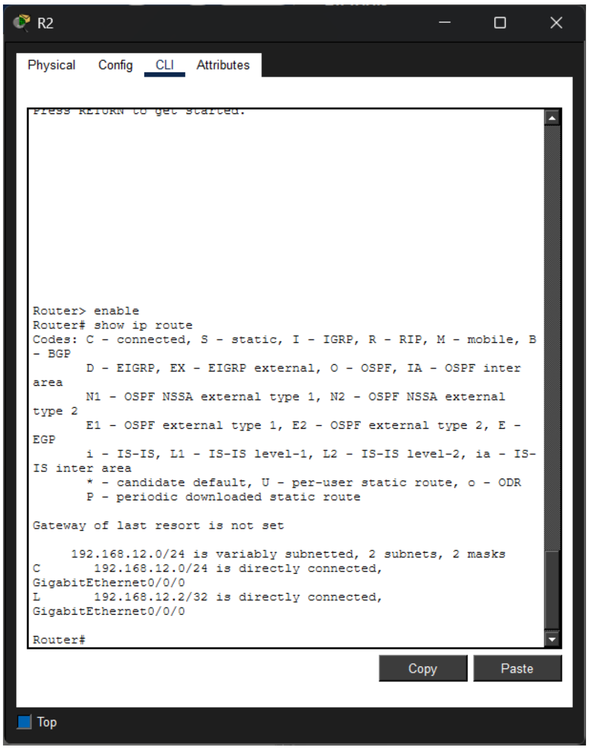
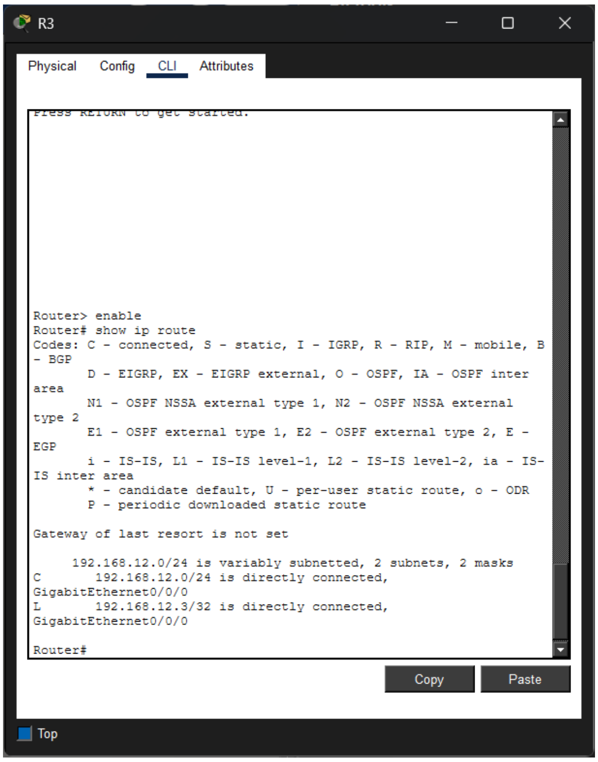
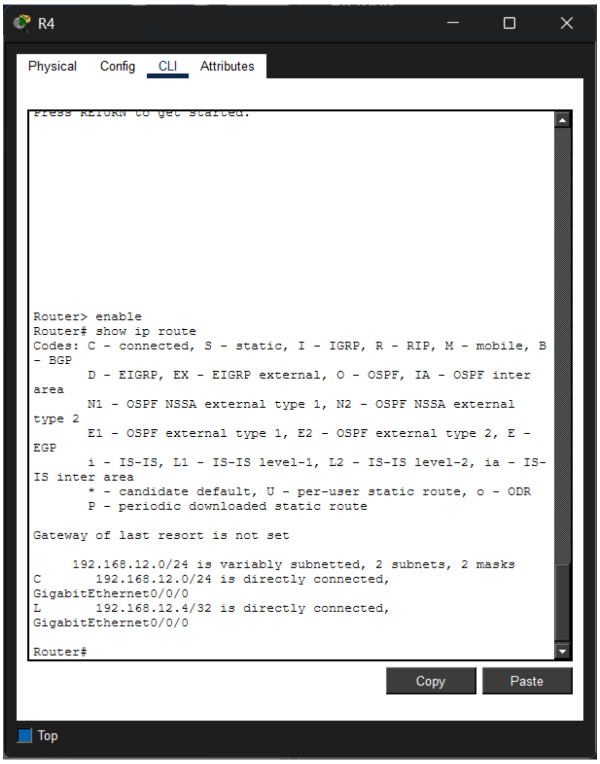
12.

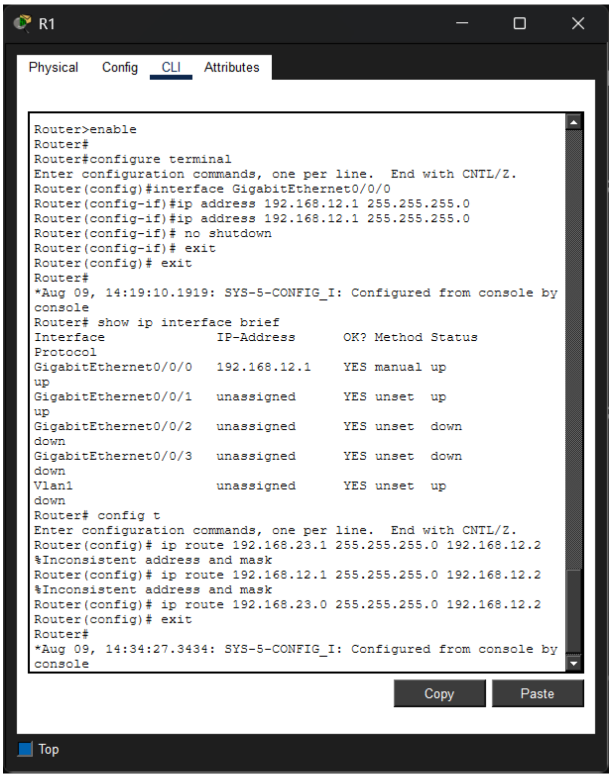




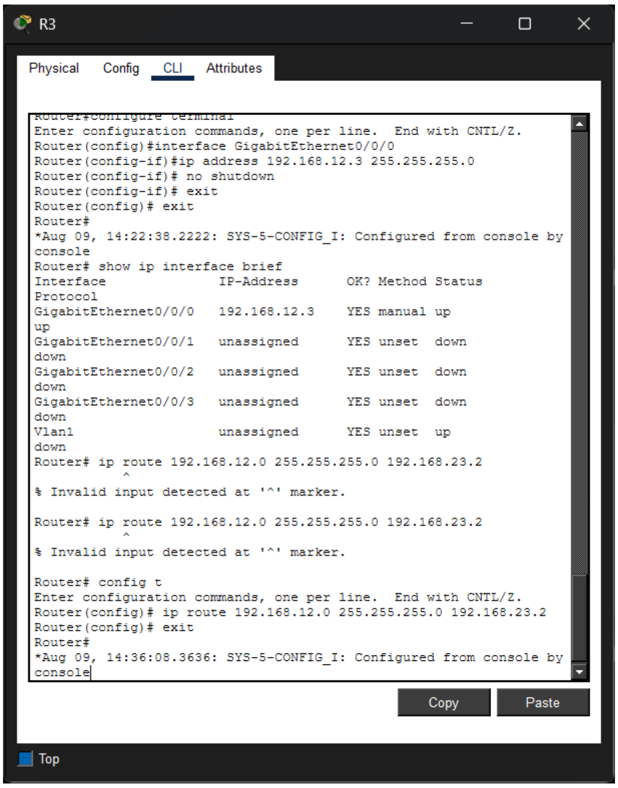




13.



```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)#ip address 192.168.12.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:19:10.1919: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.1    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  up
up
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router# config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)# ip route 192.168.23.1 255.255.255.0 192.168.12.2
%Inconsistent address and mask
Router(config)# ip route 192.168.12.1 255.255.255.0 192.168.12.2
%Inconsistent address and mask
Router(config)# ip route 192.168.23.0 255.255.255.0 192.168.12.2
Router(config)# exit
Router#
*Aug 09, 14:34:27.3434: SYS-5-CONFIG_I: Configured from console by
console
```



```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.12.3 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router#
*Aug 09, 14:22:38.2222: SYS-5-CONFIG_I: Configured from console by
console
Router# show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0/0   192.168.12.3    YES manual up
up
GigabitEthernet0/0/1   unassigned      YES unset  down
down
GigabitEthernet0/0/2   unassigned      YES unset  down
down
GigabitEthernet0/0/3   unassigned      YES unset  down
down
Vlan1                  unassigned      YES unset  up
down
Router# ip route 192.168.12.0 255.255.255.0 192.168.23.2
           ^
% Invalid input detected at '^' marker.

Router# ip route 192.168.12.0 255.255.255.0 192.168.23.2
           ^
% Invalid input detected at '^' marker.

Router# config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)# ip route 192.168.12.0 255.255.255.0 192.168.23.2
Router(config)# exit
Router#
*Aug 09, 14:36:08.3636: SYS-5-CONFIG_I: Configured from console by
console
```

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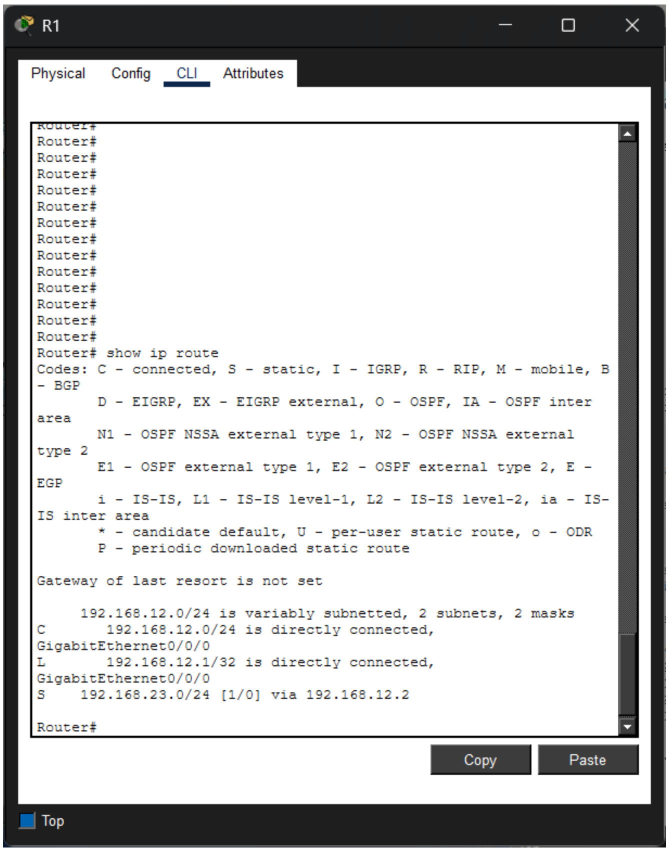
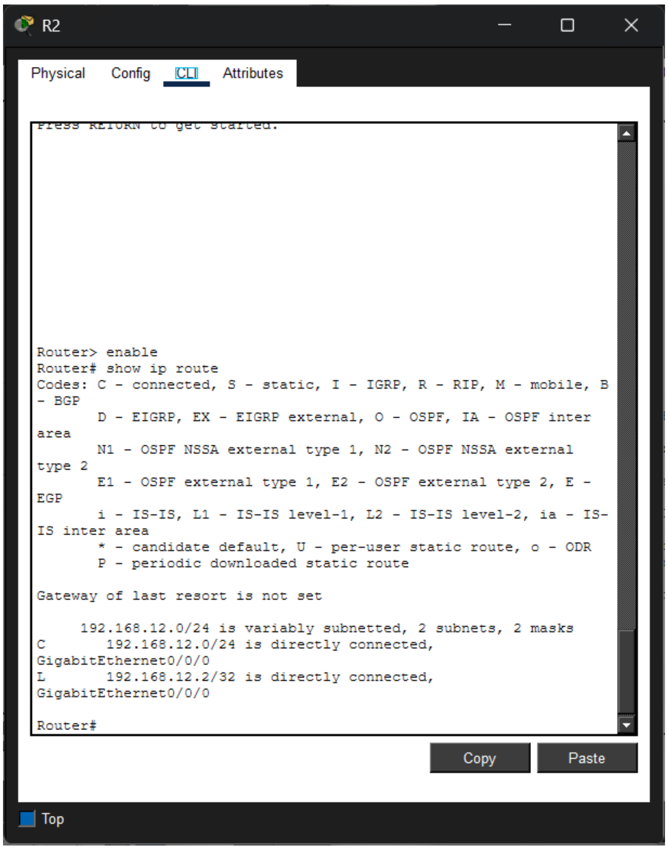
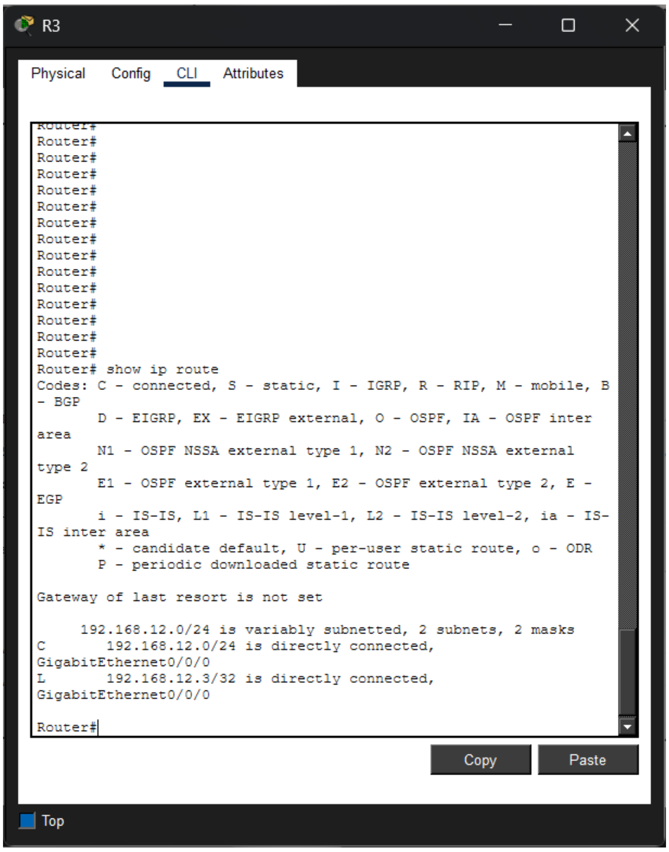
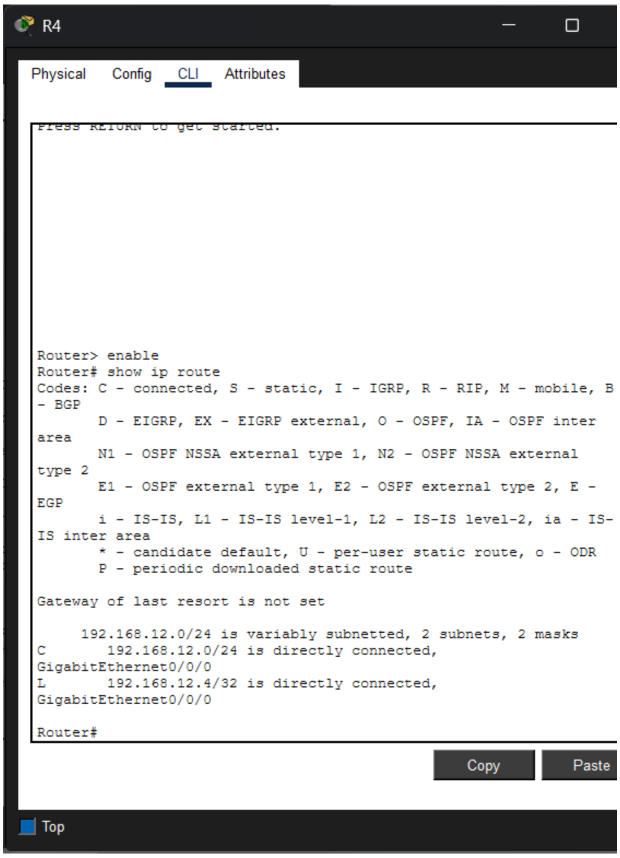
14.









15. Yes it does because it’s configured correctly.